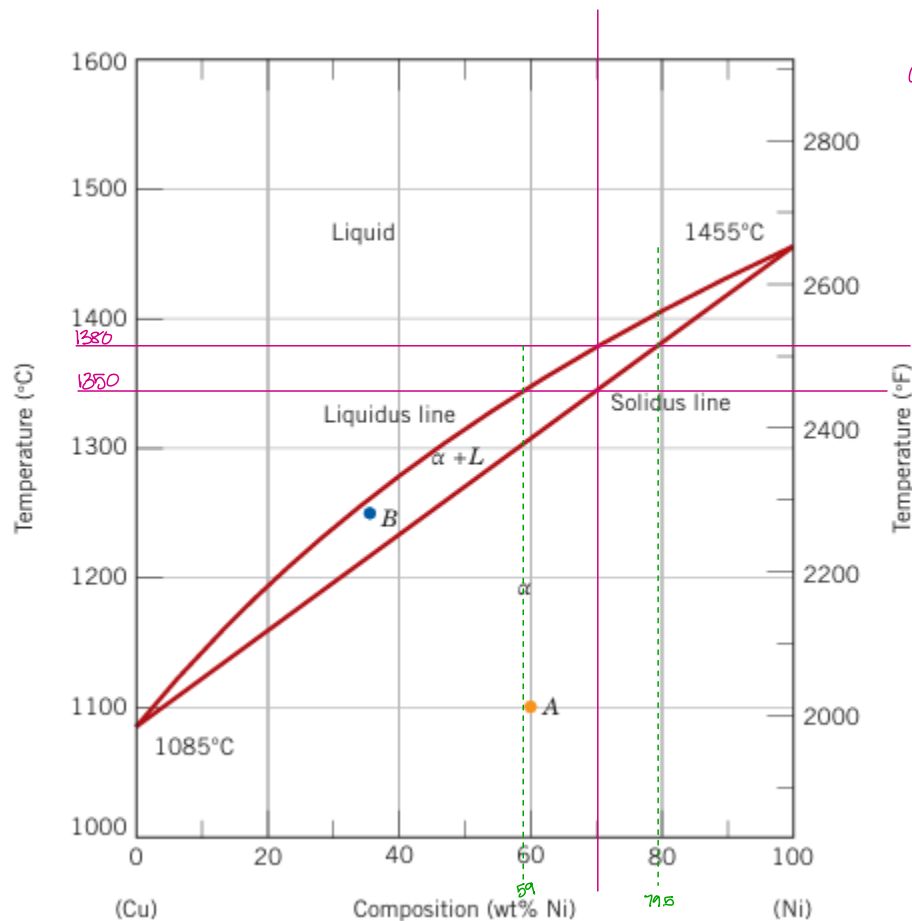


This homework is practice for the exam, so as a result, it will contain what you are expected to have in your final exam

Phase

1. A copper–nickel alloy of composition 70 wt% Ni–30 wt% Cu is slowly heated from a temperature of 1300°C (2370°F). (a) At what temperature does the first liquid phase form? (b) What is the composition of this liquid phase? (c) At what temperature does complete melting of the alloy occur? (d) What is the composition of the last solid remaining prior to complete melting? (4 points)



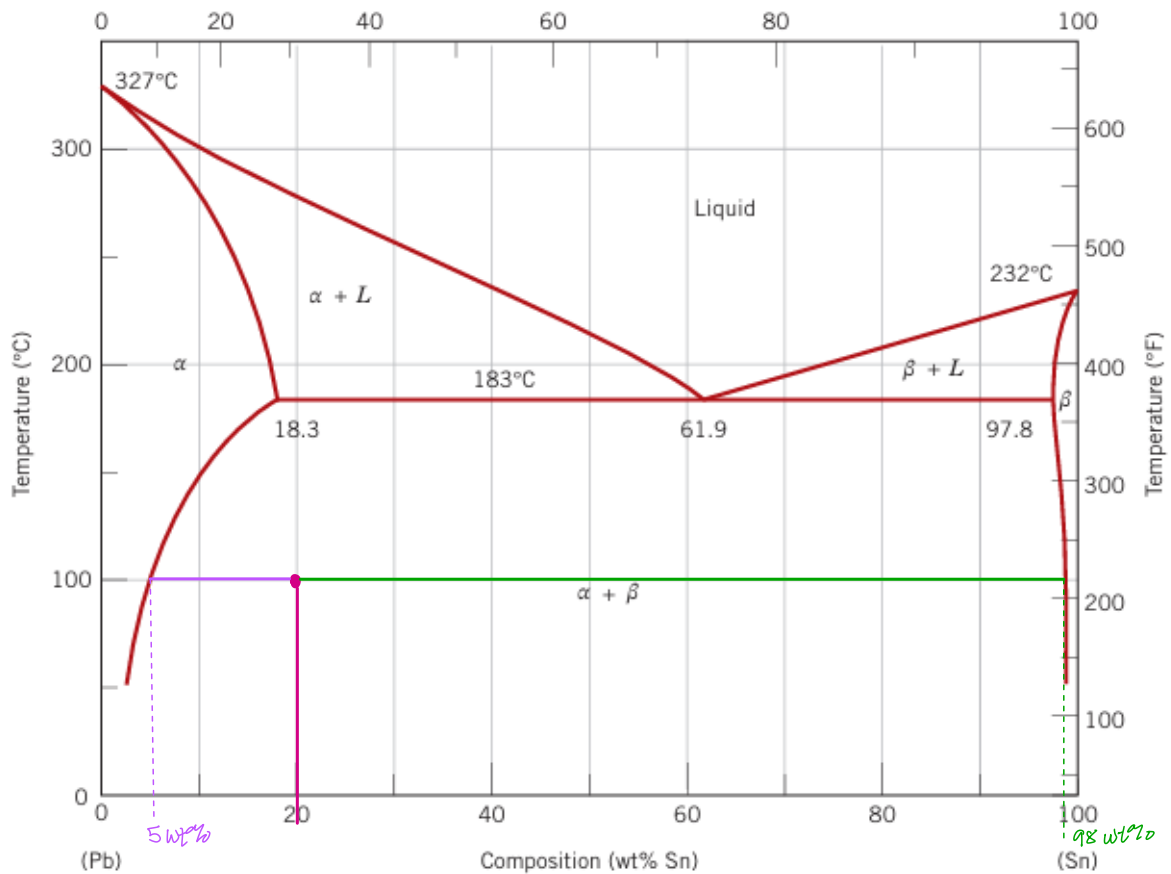
a) 1350°C

b) 59 wt% Ni

c) 1380°C

d) 79.5 wt% Ni

2. For a 20 wt% Sn–80 wt% Pb alloy at 100°C (212°F),
- What phase(s) are present?
 - What is (are) the composition(s) of the phase(s)?
 - Calculate the relative amount of each phase present in terms of mass fraction
 - Calculate the relative amount of each phase present in terms of volume fraction, taking the densities of Pb and Sn to be 11.35 and 7.29 g/cm³, respectively. (8 points)



a) solid α -Pb rich
solid β -Sn rich

b) 5 wt% α
98 wt% β

$$c) \text{wt}\% \alpha = \frac{X_B - X}{X_B - X_A} = \frac{98 - 20}{98 - 5} = \frac{78}{93} = 0.8387 = 83.87 \text{ wt}\% \alpha$$

$$\text{wt}\% \beta = \frac{X - X_A}{X_B - X_A} = \frac{20 - 5}{98 - 5} = \frac{15}{93} = 0.1613 = 16.13 \text{ wt}\% \beta$$

$$d) \rho_\alpha = \frac{1}{\frac{C_\alpha}{\rho_\alpha} + \frac{C_\beta}{\rho_\beta}} = \frac{1}{\frac{0.05}{7.29} + \frac{0.95}{11.35}} = 11.04 \frac{\text{g}}{\text{cm}^3}$$

where when $\alpha = 5 \text{ wt}\% \text{ Sn}$
when $\beta = 98 \text{ wt}\% \text{ Sn}$
Pb = 95 wt%

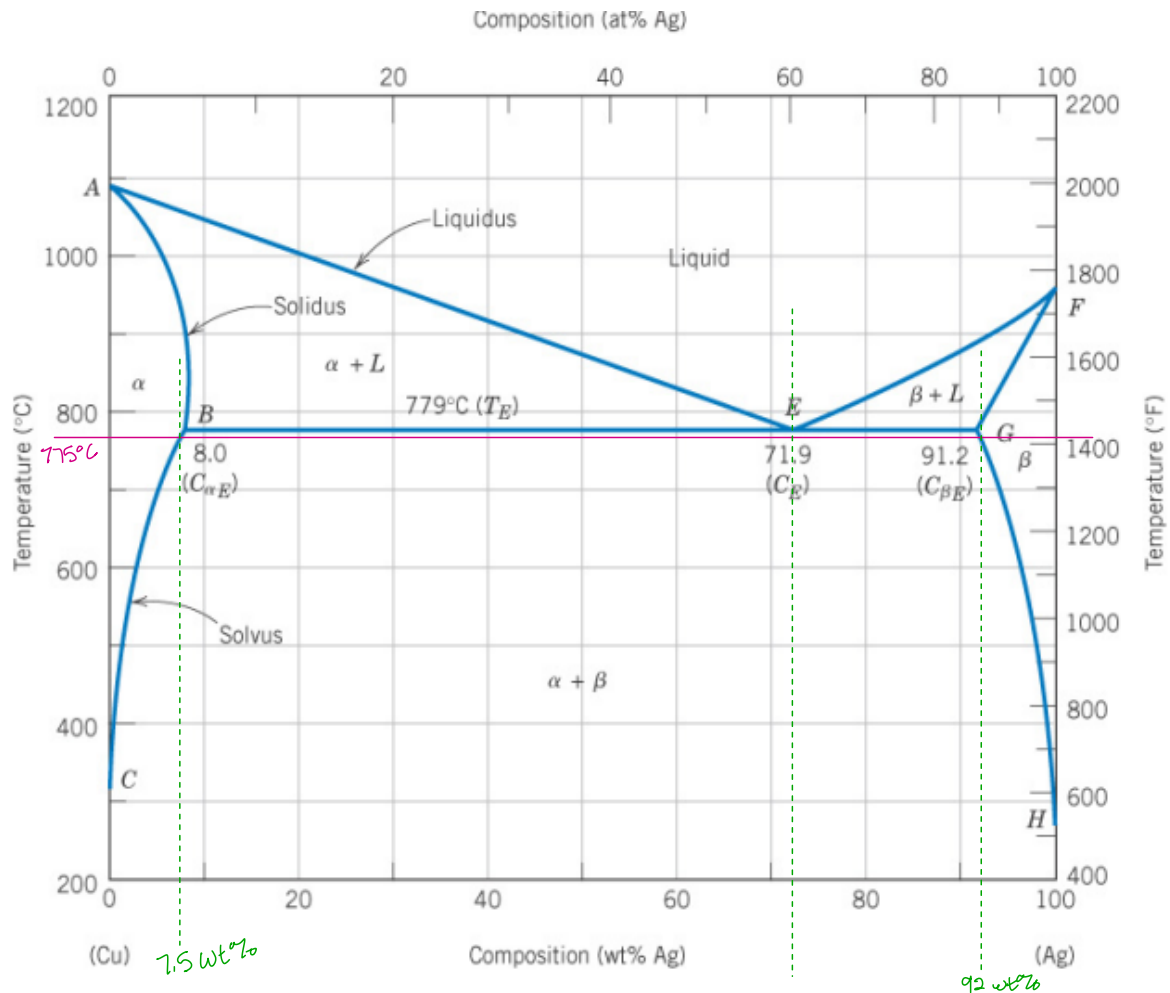
$$\rho_\beta = \frac{1}{\frac{C_\alpha}{\rho_\alpha} + \frac{C_\beta}{\rho_\beta}} = \frac{1}{\frac{0.98}{7.29} + \frac{0.02}{11.35}} = 7.24 \frac{\text{g}}{\text{cm}^3}$$

where when $\beta = 98 \text{ wt}\% \text{ Sn}$
Sn = 98 wt%
Pb = 2 wt%

$$V_\alpha = \frac{\frac{W_\alpha}{\rho_\alpha}}{\frac{W_\alpha}{\rho_\alpha} + \frac{W_\beta}{\rho_\beta}} = \frac{\frac{0.8387}{11.04 \frac{\text{g}}{\text{cm}^3}}}{\frac{0.8387}{11.04 \frac{\text{g}}{\text{cm}^3}} + \frac{0.1613}{7.24 \frac{\text{g}}{\text{cm}^3}}} = \frac{0.07595}{0.07595 + 0.021915} = 0.776$$

$$V_\beta = 1 - V_\alpha = 1 - 0.776 = 0.224$$

3. The microstructure of a Cu-Ag alloy at 775°C consists of primary α and eutectic microstructures. If the mass fractions of these two micro-constituents are 0.73 and 0.27, respectively, determine the alloy composition. (3 points)



$$\text{wt } \% \alpha = 0.73 = \frac{x_{\text{wt}} - x}{x_{\text{wt}} - x_{\alpha}} = \frac{71.9 - x}{71.9 - 7.5}$$

$$47.012 = 71.9 - x$$

$$x = 71.9 - 47.012$$

$$= 24.89 \text{ wt\% Ag} - 75.11 \text{ wt\% Cu}$$

Thermal Properties

4. a. The main span of San Francisco's Golden Gate Bridge is 1275 m long at its coldest. The bridge is exposed to temperatures ranging from -15°C to 40°C . What is its change in length between these temperatures? Assume that the bridge is made entirely of steel. (Coefficient of Linear Expansion, $\alpha = 12 \times 10^{-6} \text{ C}^{-1}$, Coefficient of Volume Expansion, $\beta = 35 \times 10^{-6} \text{ C}^{-1}$) (3 points)

$$\frac{\Delta L}{L_0} = \alpha \Delta T$$

$$\Delta L = \alpha \Delta T L_0$$

$$= (12 \times 10^{-6} \frac{1}{^{\circ}\text{C}}) (40 - (-15^{\circ}\text{C})) (1275 \text{ m})$$
$$= 0.8415 \text{ m}$$

- b. The thermal conductivity of copper is $390 \text{ W m}^{-1} \text{ K}^{-1}$. Calculate the rate of heat transfer through a copper wire with an area of 4.0 cm^2 and a length of 0.50 m . The temperature difference between the two ends of the wire is 30°C . (3 points)

$$k = \frac{390 \text{ J}}{\text{s} \cdot \text{m} \cdot \text{K}}$$

$$dT = 30 \text{ K}$$
$$dX = 0.50 \text{ m}$$

$$\frac{q}{A} = -k \frac{dT}{dX}$$

$$A = 4.0 \text{ cm}^2 \times \frac{1 \text{ m}^2}{10^4 \text{ cm}^2}$$
$$=$$

$$q = -k A \frac{dT}{dX}$$

$$= - \frac{390 \text{ J}}{\text{s} \cdot \text{m} \cdot \text{K}} (4 \times 10^{-4} \text{ m}^2) \left(\frac{30 \text{ K}}{0.50 \text{ m}} \right)$$

$$= -9.36 \frac{\text{J}}{\text{s}}$$

Electrical Properties.

5. Compute the electrical conductivity of a cylindrical silicon specimen 5.1 mm (0.2 in.) in diameter and 51 mm (2 in.) in length in which a current of 0.1 A passes in an axial direction. A voltage of 12.5 V is measured across two probes that are separated by 38 mm (1.5 in.). ($e = 1.602 \times 10^{-19} \text{ C}$) (5 points)

(a) Compute the resistance over the entire 51 mm (2 in.) of the specimen.

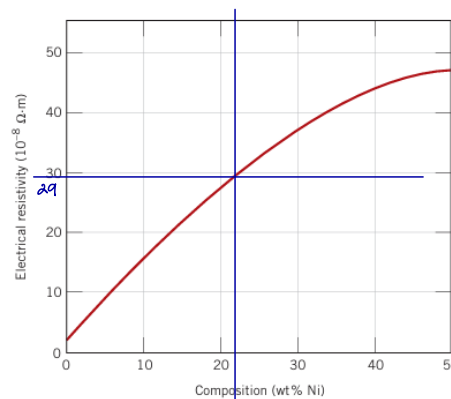
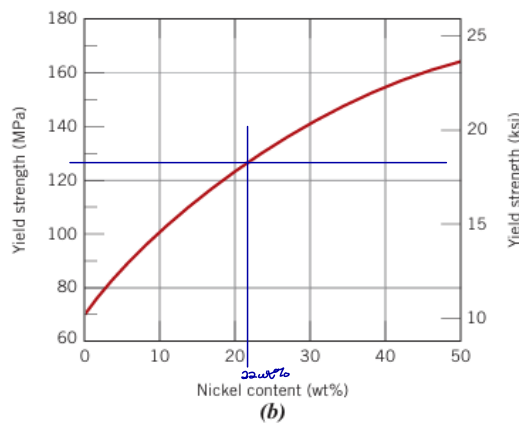
$$\sigma = \frac{I \cdot L}{V \cdot A} = \frac{(0.1 \text{ A})(0.038 \text{ m})}{(12.5 \text{ V})(\pi (\frac{0.0051 \text{ m}}{2})^2)}$$

$$= 14,88 (\Omega \cdot \text{m})^{-1}$$

$$R = \frac{L}{\sigma A} = \frac{0.051 \text{ m}}{(14,88 \Omega \cdot \text{m}) (\pi (\frac{0.0051 \text{ m}}{2})^2)}$$

$$= 167.8 \Omega$$

(b) Determine the electrical conductivity of a Cu-Ni alloy that has a yield strength of 125 MPa (18,000 psi).



$$\sigma = \frac{1}{\rho} = \frac{1}{29 \times 10^{-8} (\Omega \cdot \text{m})}$$

$$= 3.45 \times 10^6 (\Omega \cdot \text{m})^{-1}$$

6. For intrinsic gallium arsenide, the room-temperature electrical conductivity is $3 \times 10^{-7} (\Omega \cdot \text{m})^{-1}$; the electron and hole mobilities are, respectively, 0.80 and $0.04 \text{ m}^2/\text{V} \cdot \text{s}$. Compute the intrinsic carrier concentration n_i at room temperature. (5 points)

$$\sigma = n_i e (\mu_e + \mu_h)$$

$$n_i = \frac{\sigma}{e (\mu_e + \mu_h)}$$

$$= \frac{3 \times 10^{-7} (\Omega \cdot \text{m})^{-1}}{1.60 \times 10^{-19} \text{ C} (0.84 \frac{\text{m}^2}{\text{V} \cdot \text{s}})}$$

$$= 2.23 \times 10^{12} \text{ m}^{-3}$$

$$\sigma = 3 \times 10^{-7} (\Omega \cdot \text{m})^{-1}$$

$$\mu_e = 0.80 \frac{\text{m}^2}{\text{V} \cdot \text{s}}$$

$$\mu_h = 0.04 \frac{\text{m}^2}{\text{V} \cdot \text{s}}$$

$$e = 1.60 \times 10^{-19} \text{ C}$$

Magnetic Properties

7. The magnetic flux density within a bar of some material is 0.435 tesla at an H field of $3.44 \cdot 10^5 \text{ A/m}$. Compute the following for this material: (a) the magnetic permeability, and (b) the magnetic susceptibility. (c) What type(s) of magnetism would you suggest is displayed by this material? Why? $\mu_0 = 4\pi \times 10^{-7} \text{ T} \cdot \text{m/A}$ (6points)

$$B = 0.435 \text{ T}$$

$$H = 3.44 \times 10^5 \text{ A/m}$$

$$\mu_0 = 4\pi \times 10^{-7} \frac{\text{T} \cdot \text{m}}{\text{A}}$$

$$a) \mu = \frac{B}{H} = \frac{0.435 \text{ T}}{3.44 \times 10^5 \text{ A/m}} = 1.26 \times 10^{-6} \frac{\text{T} \cdot \text{m}}{\text{A}}$$

$$b) \chi_m = \frac{\mu}{\mu_0} - 1 = \frac{1.26 \times 10^{-6}}{4\pi \times 10^{-7}} - 1 = 2.1676 \times 10^{-3}$$

c) The χ_m is small & positive which is characteristic of paramagnetic materials where the magnetic moments align slightly with the external field

Optical Properties

8. a. Visible light having a wavelength of 6×10^{-7} m appears orange. Compute the frequency and energy of a photon of this light. ($c = 3 \times 10^8$ m/s, $h = 6.626 \times 10^{-34}$ J.s) (4 points)

$$c = \lambda \nu$$
$$\nu = \frac{c}{\lambda} = \frac{3 \times 10^8 \frac{\text{m}}{\text{s}}}{6 \times 10^{-7} \text{ m}} = 5 \times 10^{14} \frac{1}{\text{s}}$$
$$E = h\nu = (6.626 \times 10^{-34} \text{ J.s}) (5 \times 10^{14} \frac{1}{\text{s}})$$
$$= 3.313 \times 10^{-19} \text{ J/photon}$$

$$= \frac{3.313 \times 10^{-19} \text{ J}}{1.602 \times 10^{-19} \text{ J/eV}} = 2.07 \text{ eV}$$

Ceramics

9. You are evaluating the mechanical stability of brittle ceramic catalyst supports for a packed-bed reactor. To determine their flexural strength (σ), you perform a center-point loading test with a support span of $L = 18$ mm. (6 points)

You test two different batches of **rectangular** pellets. Both have a width $b = 6.5$ mm and a depth $d = 7.5$ mm.

- Batch A fractures at an applied force $F = 120$ N.
- Batch B fractures at an applied force $F = 145$ N.

Calculate the flexural strength (σ) in MPa for both batches. Which formulation is stronger?

$$\sigma_A = \frac{3F_AL}{2bd^2} = \frac{3(120\text{ N})(0.018\text{ m})}{2(0.0065\text{ m})(0.0075\text{ m})^2} = 8.86 \times 10^6 \text{ Pa} = 8.86 \text{ MPa}$$
$$\sigma_B = \frac{3(145\text{ N})(0.018\text{ m})}{2(0.0065\text{ m})(0.0075\text{ m})^2} = 10.71 \text{ MPa}$$

$$L = 18 \text{ mm} = 0.018 \text{ m}$$
$$b = 6.5 \text{ mm} = 0.0065 \text{ m}$$
$$d = 7.5 \text{ mm} = 0.0075 \text{ m}$$

Batch B is the stronger formulation

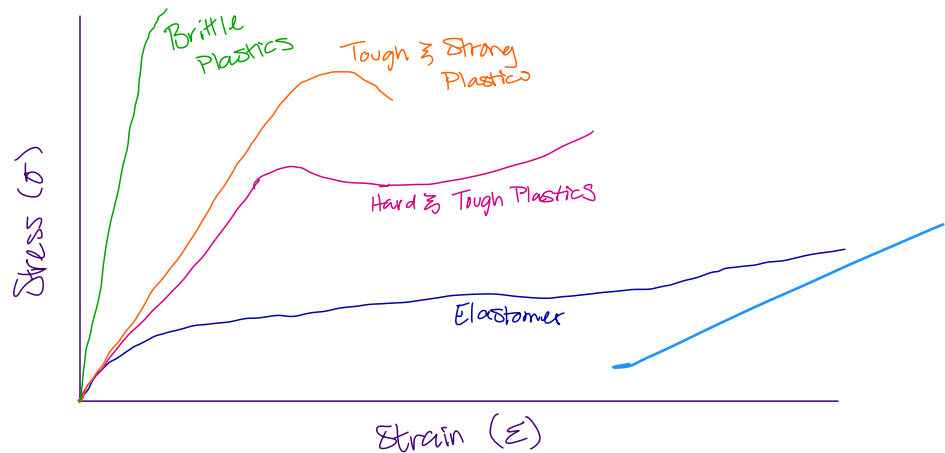
Polymer

- a. Which has a slower conversion, chain growth or step growth formation? (Hint: draw the graph) (2 points)

Step growth formation is a slow addition of smaller oligomers reacting with each other to build chains.

Chain growth is a radical reaction which tends to be very reactive and quick. There is a loss of control on the length of the chain for this process.

- b. Sketch a stress-strain curve to highlight the behavior of brittle polymers, plastics, and elastomers during tensile testing (2 points)



Biomolecules

10. A. What are four types of biomacromolecules? (2 points)

Proteins
Nucleic Acids
Carbohydrates
Lipids

b. What specific type of chemical reaction links amino acids, monosaccharides, and nucleotides into their respective polymers? What small molecules are always released as a byproduct of this reaction? (2 points)

Condensation. The linking of the amino acids produces water molecules

Nanomaterial

11. a. What are the classifications of nanomaterials (2 points)

0 dimension: nanoparticles
1 dimension: nanowires/tubes
2 dimension: nanosheets/films
3 dimension: bulk/nanocrystalline

B. A researcher wants to deposit a precisely controlled, ultrathin oxide film on a silicon substrate one atomic layer at a time, then verify the elemental composition of the deposited film. (a) Which synthesis technique (top-down or bottom-up) and specific method would be most appropriate, and why? (b) Which characterization technique would best confirm the elemental composition and bonding nature of the resulting film? (3 points)

a) The bottom up approach. Since precise control and not a lot of it is needed, a bottom up technique like Atomic-Layer Deposition (ALD) would be ideal

b) X-ray Photoelectron Spectroscopy (XPS) is best for confirming elemental composition and chemical bonding states of the film.